# Narrative Chain-of-Thought: Inference-Time Scaffolding for Defeasible Ethical Reasoning in Large Language Models

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## Abstract

Standard chain-of-thought on moral dilemmas exhibits two structural failure modes that matter for deployment: stakeholder collapse, where the model reduces a multi-party situation to a binary, and uncertainty suppression, where the model commits to a projected outcome it cannot warrant. We introduce narrative chain-of-thought (N-CoT), a system prompt that asks the model to name a protagonist, enumerate stakeholders, project two-step consequences, articulate uncertainty, and only then commit. The protocol adds no training data, parameters, or fine-tuning. On a 100-scenario DailyDilemmas sample across four generators from three vendors, N-CoT cuts stakeholder collapse from up to 31% to under 1% and uncertainty suppression from up to 72% to 1–24% on every model. A section-by-section ablation attributes each structural shift to the specific sub-instruction that carries it, with negligible cross-variable spillover. Extended to a four-round multi-stakeholder protocol with a moderator-built integrated proposal and a binary accept/reject vote, the same scaffold converts a 6% debate standoff into 95% full consensus while preserving a small fraction of principled structural rejections in the roles whose stake the integrated proposal materially undermines. The intervention is a small, auditable change to the prompt that removes two failure modes firing near-universally on current frontier models.

## 1 Introduction

A frontier LLM asked to think step by step about an everyday moral dilemma frequently produces a trace that does not look like ethical reasoning. On the DailyDilemmas corpus (Chiu et al., 2025), four frontier generators spanning three vendors (OpenAI, Anthropic, xAI) and two model tiers suppress articulated uncertainty on 50–72% of standard-CoT outputs and collapse the stakeholder set on 15–31%. This is a structural deficit of the trace itself: the model commits to a projected outcome it cannot warrant, and it does so against a flattened picture of who is affected. The same shape of deficit is coherent with deployment-relevant failures reported elsewhere, including the GPT-4o sycophancy rollback (OpenAI, 2025b,a), agentic-misalignment blackmail rates up to 96% in simulated deployments (Lynch et al., 2025), and reasoning-induced misalignment (Yan et al., 2025; Sharma et al., 2024).

We introduce narrative chain-of-thought (N-CoT), a system prompt that puts the model on the manifold of human narrative its pretraining corpus most heavily samples. The model is asked, in first person, to name a protagonist, enumerate stakeholders, project two-step consequences, articulate uncertainty, and only then commit. Figure 1 states the mechanism. The methodological move is the one Gottweis et al. (2025) make for scientific discovery and Lu et al. (2024) extend to end-to-end research workflows: scaffold LLM reasoning into the primitives of the target domain. The primitives we reify are the primitives of human ethical reflection. At the multi-agent layer we replace the AI co-scientist's meta-review role with a moderator that integrates competing stakeholder positions into a single proposal and asks each stakeholder to vote (§5).

This paper reports three results. First, on a 100-scenario DailyDilemmas sample across the four-generator quartet, N-CoT cuts both failure modes to floor on every model (§4). Second, a sub-instruction ablation on `claude-sonnet-4-6` attributes each structural shift to the specific N-CoT sub-instruction that carries it, with negligible cross-variable spillover (§4.1). Third, extended to a four-round multi-stakeholder protocol with an integrated-proposal vote, the scaffold converts a 6% debate standoff into 95% full consensus

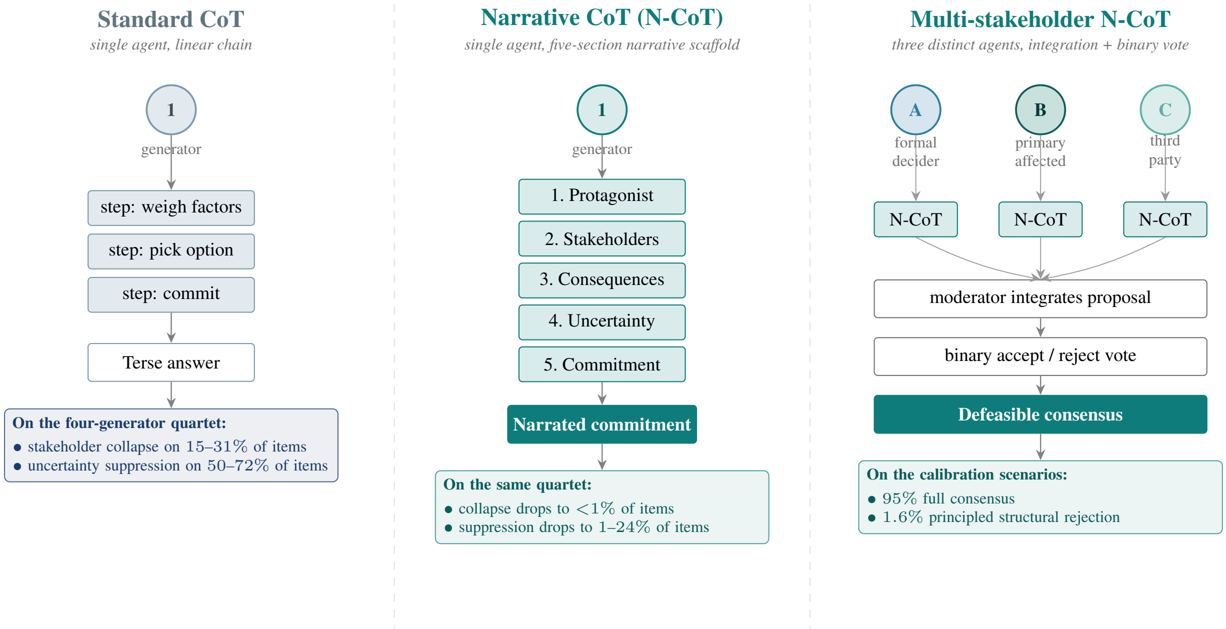


Figure 1: One scaffold, three settings. Standard CoT (left) lets one generator run a linear reasoning chain over the dilemma; the two failure modes diagnosed in §4 fire on this scaffold. Narrative CoT (middle) keeps the single generator but forces the trace through five narrative primitives (protagonist, stakeholders, consequences, uncertainty, commitment) before any commit; on the four-generator quartet this drops stakeholder collapse to $<1\%$ and uncertainty suppression to 1–24% of items. Multi-stakeholder N-CoT (right) reuses N-CoT as the per-agent generator and adds three distinct stakeholder narrators (formal decider, primary affected party, third party) whose modification requests are integrated by a moderator and then put to a binary accept/reject vote; on the calibration scenarios this protocol yields 95% full consensus with 1.6% principled structural rejection (§5). The same scaffold composes from one narrator to many.

while preserving a small 1.6% structural-rejection rate concentrated in roles whose stake the integrated proposal undermines (§5). The intervention is a small auditable prompt change. It does not solve alignment, and it does not by itself reduce the saturated downstream deployment probes that motivated the diagnosis (Appendix E). What it does is remove two failure modes that fire near-universally on current frontier models, and produce defeasible multi-agent commitments that revise under counter-evidence yet hold firm where revision would harm a stakeholder.

## 2 Background and Related Work

**Chain-of-thought and its failure modes.** Chain-of-thought prompting (Wei et al., 2022; Kojima et al., 2022) elicits step-by-step intermediate reasoning before a final answer; it improves arithmetic and symbolic tasks but does not by construction force the trace to track situational structure. Recent analysis identifies a sharp tension: as chain-of-thought reasoning capacity grows, models become more responsive to harmful requests because reasoning and safety entangle in shared neuron populations, a phenomenon Yan et al. (2025) describe as "reasoning-induced misalignment". N-CoT can be read as a constrained chain-of-thought that re-anchors the trace to a narrative-causal structure before the reasoning capacity is exercised; §4 shows that the intervention works on a current reasoning model as well as a non-reasoning one.

**Narrative scaffolding for reasoning.** Visualisation-of-Thought (Wu et al., 2024) shows that asking a model to externalise reasoning in the native representational format of a domain (spatial diagrams for spatial tasks) improves reliability. We apply the same principle to ethical reasoning: the native format is a story with named agents, projected consequences, and uncertainty (MacIntyre, 1981; Bruner, 1986). Empirical narrative-perspective interventions in medical decision contexts measurably increase perspective-taking and empathic concern (Shaffer et al., 2019; Bientzle et al., 2021, 2024), and recent work on generative agents shows long-horizon coherent behaviour is achievable from narrated character state (Park et al., 2023).

**Causal inference from narrative.** The Computational Theory of Mind (Fodor, 1975; Putnam, 1967) treats understanding a sequence of events as inferring a structural causal model that explains them; the Kuleshov experiment (Kuleshov, 1974; Bordwell, 1985) is a textbook example. LLMs perform near random baseline on causal inference stripped of empirical pattern matches (Jin et al., 2024): pretraining supplies the ability to pattern-match narrative token sequences, not the structured causal reasoning that determines how those narratives resolve. The role of N-CoT is to convert that pattern matching into a usable causal trace at inference time.

**Multi-agent debate and defeasibility.** Irving et al. (2018) proposed AI safety via debate; Du et al. (2023) showed multi-agent debate improves factual reasoning. Both target task-correctness, not stakeholder-divergent value standoffs. Defeasible reasoning, the property that conclusions can be revised in light of new information without abandoning the underlying inference scheme, has a long lineage in moral philosophy (Pollock, 1987; MacIntyre, 1981). The protocol we evaluate in §5 stacks defeasibility on top of N-CoT and asks whether present-day models can be elicited to revise narrated commitments under narrated counter-evidence.

**Multi-agent LLM systems for scientific discovery.** A growing line of work scaffolds LLM reasoning into the primitives of a target domain. The AI co-scientist of Gottweis et al. (2025) reifies hypothesis generation, peer review, ranking, evolution, and meta-review as co-operating agents; the AI Scientist (Lu et al., 2024) closes the loop end-to-end; Coscientist (Boiko et al., 2023), ChemCrow (M. Bran et al., 2024), Agent Laboratory (Schmidgall et al., 2025), and the Virtual Lab (Swanson et al., 2025) instantiate the move in chemistry and biomedicine. Our protocols ask the analogous question one level down: can a scaffold that reifies the primitives of human ethical reflection deliver the same kind of inference-time quality lift on moral reasoning that scientific-method scaffolds deliver on hypothesis generation? The moderator's integrated-proposal step in §5 is structurally the meta-review of Gottweis et al. (2025), applied to value-laden positions.

**Algorithm 1** Narrative chain-of-thought (N-CoT): generation and structural coding for a single dilemma.

**Require:** dilemma $s$; generator $G$; judge $J$; decoder $\Theta$
**Ensure:** trace $t$ and structural codes $\mathbf{c}$

- **Build N-CoT system prompt** $\pi_{\text{NCOT}}$ requesting five first-person sections:
  - (1) PROTAGONIST: name role and epistemic state
  - (2) STAKEHOLDERS: parties intersecting the decision
  - (3) CONSEQUENCES: each action $\geq 2$ steps forward
  - (4) UNCERTAINTY: what remains genuinely unknown
  - (5) COMMITMENT: chosen action and narrative warrant
- **Generate trace**
  - $t \leftarrow G(\pi_{\text{NCOT}} \parallel s; \Theta)$
- **Code structural variables** with judge $J$'s rubric
  - $(sc, mh, us, nf, ref, tr) \leftarrow J(t)$
  - *// stakeholder count, max causal hops, uncertainty score,*
  - *// framework count, refused flag, truncated flag*
- **Derive failure-mode tags** (deterministic)
  - STAKEHOLDERCOLLAPSE $\leftarrow sc \leq 1$
  - UNCERTAINTYSUPPRESSION $\leftarrow us = 0$
  - CONSEQUENTIALFLATTENING $\leftarrow mh \leq 1$
  - FRAMEWORKENUMERATION $\leftarrow nf \geq 2$
  - PREMATUREREFUSAL $\leftarrow ref$
- **return** $(t, \mathbf{c})$

## 3 Narrative Chain-of-Thought

Narrative chain-of-thought (N-CoT) is a system prompt that instructs the model to produce, in first person, a five-section trace before any final answer: (1) characterise the protagonist (name, role, what they know); (2) enumerate the stakeholders whose lives intersect the decision and what is at stake for each; (3) project the consequences of each available action at least two steps out for each stakeholder; (4) state what remains uncertain about each projected future; (5) commit to a decision and explain why, within the narrative frame, that trajectory is preferable to the alternatives.

The intervention is a single change to the system prompt; it adds no training data, parameters, or fine-tuning, and it does not mention causal graphs, utilities, or any formal apparatus. The mechanism it exploits is that text matching the five-section structure is densely represented in the narrative subdistribution of the pretraining corpus, so the model can amortise causal-trajectory inference by retrieval rather than by reasoning from first principles. Algorithm 1 states the procedure.

### 3.1 Multi-Stakeholder Narrative Deliberation

A single N-CoT trace commits one protagonist to a position; many real decisions involve multiple narrators with conflicting stakes. We extend

N-CoT into a four-round protocol that re-uses N-CoT as the per-agent generator. For each scenario, three stakeholder perspectives (formal decider, primary affected party, third party) each produce an N-CoT statement (Round 0), exchange rebuttals (Round 1), restate a final position (Round 2), respond to a moderator-built synthesis with `ACCEPT`, `ACCEPT_WITH_MODIFICATION`, or `REJECT` (Round 3), and cast a binary `ACCEPT/REJECT` vote on a single integrated proposal the moderator constructs from the three modification requests (Round 4). The integration step in Round 4 makes defeasibility measurable: an agent who continues to reject a proposal that addresses its modifications is signalling a structurally non-negotiable concern, not procedural friction.

## 4 Experiment 1: N-CoT at Scale

**Setup.** Three prompting conditions are compared: bare input/output, standard CoT ("Think step by step, then give your answer"), and N-CoT (the five-section structure of §3, with no formal apparatus mentioned). Decoding parameters are held identical across conditions. Four generators are tested: `gpt-5.4-nano` ($N$=20) from the original pilot, plus three new generators spanning two additional vendors: `claude-haiku-4-5` ($N$=20), `grok-4-1-fast-reasoning` ($N$=5), and the flagship `claude-sonnet-4-6` ($N$=5, cost-controlled). Here $N$ is the number of independent samples per (scenario, condition, generator) cell drawn at temperature 1; with the 100-scenario sample and three conditions this yields $300N$ generations per generator before the cross-judge coding stage. Coded sample sizes ($N_{\mathrm{S}}, N_{\mathrm{N}}$ in Table 1) are slightly smaller because the rubric drops generations where the judge could not extract a coded variable. Outputs are coded by two cross-vendor judges (`claude-haiku-4-5` primary, `gpt-5.4-nano` secondary) on a six-variable rubric with quadratic-weighted Cohen's $\kappa$ (Cohen, 1960) per variable. The scenario pool is a 100-scenario stratified sample of DailyDilemmas (Chiu et al., 2025), spanning 18 topic groups; the same deterministic stratified sample (seed 42) is used for all four generators, making the scenario sets comparable.

**N-CoT eliminates both failure modes on every model.** Uncertainty suppression fires on 50.0–72.3% of standard-CoT outputs across the quartet (Table 1); stakeholder collapse on 14.6–30.6%.

| Generator (vendor, tier) | $N_{\mathrm{S}}$ | $N_{\mathrm{N}}$ | $\Delta_{\mathrm{SC}}$ | $\Delta_{\mathrm{US}}$ |
|---|---|---|---|---|
| gpt-5.4-nano (OpenAI, b) | 1,989 | 1,985 | −30 | −72 |
| claude-haiku-4-5 (Anth., b) | 1,999 | 1,996 | −25 | −29 |
| grok-4-1-fast-r. (xAI, b) | 515 | 515 | −13 | −44 |
| claude-sonnet-4-6 (Anth., f) | 500 | 500 | −26 | −37 |

Table 1: Cell sizes and percentage-point drops for the two failure modes that empirically fire ($\Delta_{\mathrm{SC}}$ = stakeholder collapse, $\Delta_{\mathrm{US}}$ = uncertainty suppression; both N-CoT minus standard CoT, in percentage points; $N_{\mathrm{S}}, N_{\mathrm{N}}$ are coded samples under standard CoT and N-CoT respectively). Vendor tiers are b = budget, f = flagship. Figure 2 plots the raw rates with the same drop annotations. Premature refusal, framework enumeration, and consequential flattening fire below 1% on all generators and are reported as untested.

Both collapse under N-CoT: stakeholder collapse to 0.0–1.2%, uncertainty suppression to 0.8–24.5% (Figure 2). The reduction is largest where the standard-CoT rate is largest, and the pattern is cross-vendor: no single vendor escapes the baseline, and no single vendor escapes the intervention. `gpt-5.4-nano` collapses both modes to <1%. The two Anthropic generators retain a 20–25% residual on uncertainty suppression that the prompt-side intervention does not reach. The three additional hypothesised failure modes (premature refusal, framework enumeration, consequential flattening) fire below 1% across all cells under either condition and are reported as untested.

**The effect on rates is matched by a large effect on the underlying structural metrics.** N-CoT outputs carry positive Cliff's $\delta$ (Cliff, 1993) on stakeholder count (+0.48 to +0.99 across the quartet) and uncertainty score (+0.63 to +0.99). N-CoT outputs are also 5–7× longer than standard-CoT outputs, so length is the parsimonious null. After regressing each metric on $\log(\text{output length})$ pooled across conditions and re-evaluating $\delta$ on residuals, the structural shift survives length removal on the OpenAI and xAI generators (`gpt-5.4-nano` $\delta_{\text{resid}} = +0.51$ on stakeholder count and +0.77 on uncertainty score; `grok-4-1-fast-reasoning` +0.33 and +0.43; all CIs strictly above zero). On the two Anthropic generators the residualised effect is statistically indistinguishable from zero, which we read as those models already exhibiting the target structural density per unit length under standard CoT, so the prompt-side gain there operates through additional length rather than additional per-unit density. Per-

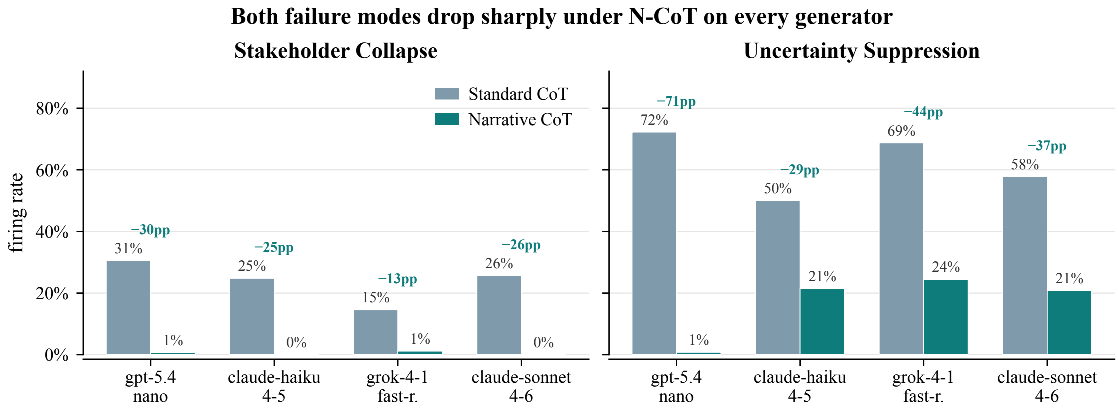


Figure 2: Failure-mode firing rates (standard CoT vs. N-CoT) across the four-generator quartet (`gpt-5.4-nano`, `claude-haiku-4-5`, `grok-4-1-fast-reasoning`, `claude-sonnet-4-6`). The suppression pattern from the original pilot replicates across vendors and model tiers; uncertainty suppression drops by 28–72 percentage points and stakeholder collapse drops to near zero on every model.

generator $\delta$s with bootstrap CIs are in Appendix B (Table 5, Figure 5).

### 4.1 Sub-Instruction Ablation

To identify which section of the N-CoT prompt carries the structural shift, we run a sub-instruction ablation on `claude-sonnet-4-6` ($N$=3, 30-scenario stratified subsample, seed 43; cost-controlled). Six conditions are compared: the full N-CoT prompt (control) and five variants each dropping exactly one of the five sections. Figure 3 reports Cliff's deltas for each drop condition relative to the full N-CoT control on the two headline structural variables.

The ablation shows direct causal attribution. Dropping Stakeholders reduces stakeholder count by $\delta = -0.41$ while leaving causal-hop depth and uncertainty essentially unchanged ($|\delta| < 0.03$); dropping Uncertainty reduces uncertainty score by $\delta = -0.92$, nearly the maximum possible, while leaving every other variable untouched; dropping Consequences reduces causal-hop depth by $\delta = -0.22$ with small spillover onto stakeholder count, consistent with multi-step consequence projection forcing the model to name the entities those consequences fall on. Dropping Protagonist or Commitment contributes no signed effect on any structural variable. The diagonal pattern is the signature of section-level mechanism: a single emergent narrative factor would dilute every variable uniformly when any section is dropped, and a pure length confound would shrink the forward-window column most regardless of which section is dropped. Neither fires. Each substantive sub-instruction carries the structural variable the main paper attributes to it and little else (full table: Appendix F, Table 9).

## 5 Experiment 2: Multi-Stakeholder Narrative Deliberation

**Setup.** Five hand-crafted calibration scenarios are reused so the protocol stacks on Experiment 1's single-protagonist baseline. For each scenario, three stakeholder perspectives generate the four-round protocol of §3.1, $N$=10 samples per cell, on the `gpt-5.4-nano` pilot. Round 0 reuses cached N-CoT statements. The moderator is `gpt-4o-mini`. A cross-vendor moderator robustness check using `claude-sonnet-4-6` over the same cached agent traces is reported in Appendix A and yields a statistically indistinguishable consensus rate.

**Progressive structuring converts a categorical standoff into near-universal consensus.** Figure 4 reports the convergence arc across four protocol designs. With a closed action taxonomy and three rounds of pure debate, only 6% of debates reach full consensus. Opening the action space and authorising the moderator to surface synthesis raises consensus to 9% while revealing the underlying mechanism: agents propose novel actions in 73% of final-round outputs and a coherent moderator-built synthesis emerges in 82% of debates, but the agents do not self-coordinate onto those syntheses. Presenting the synthesis back to the agents (Round 3) drives outright rejection to 0% and partial convergence ($\geq 2/3$ agreement) to 100%, while full consensus stays at 0% because every agent demands a different modifica-

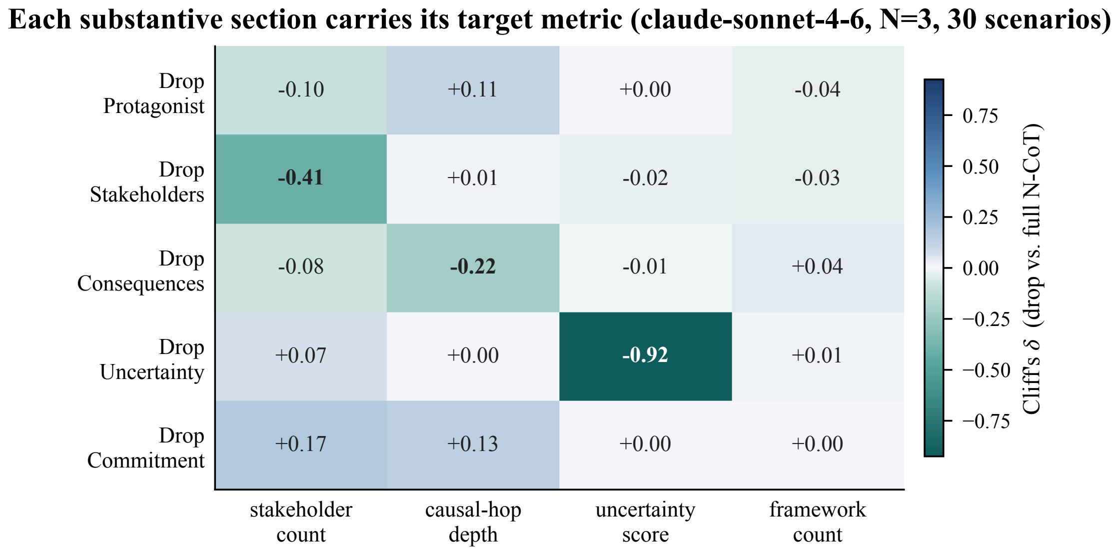


Figure 3: Sub-instruction ablation on `claude-sonnet-4-6`: Cliff's $\delta$ for each drop-one-section condition vs. the full N-CoT control on stakeholder count and uncertainty score. Negative $\delta$ means the section raised the structural variable, so dropping it reduces it. The Stakeholders section (Section 2) and the Uncertainty section (Section 4) show the largest negative $\delta$ on their respective target variables.

tion. Integrating those modifications into a single proposal and forcing a binary accept/reject vote (Round 4) produces $78/82 = 95.1\%$ full consensus and $242/246 = 98.4\%$ individual acceptance (Wilson 95% CIs $[88.1, 98.1]\%$ and $[95.9, 99.4]\%$), with the integrated proposal addressing a mean 2.98 of the 3 agent modifications per debate. From 5 scenarios $\times 2$ generators $\times 10$ samples $= 100$ designed debates, 82 produced a synthesis and proceeded to Round 4, yielding $82 \times 3 = 246$ individual binary votes.

**The residual rejections are structurally interpretable.** The vote produces 4 rejections out of 246 (1.6%, Wilson $[0.6, 4.1]\%$). All four concentrate in the `primary_affected` and `third_party` agent roles, on debates where honouring one stakeholder's modification structurally undermines another's stake (`pharma_whistleblower` on the senior colleague role, `av_engineer` on the future-pedestrian role). This is the structural shape a defeasibility-respecting protocol should produce. With $N{=}4$ rejections the qualitative pattern is suggestive rather than statistically confirmatory, and the full-quartet replication is pre-registered as the actual test.

**Scope.** The convergence arc is established on the `gpt-5.4-nano` pilot; replicating the four-round protocol on the full quartet would require ∼10,000 additional long-context generations and is deferred. Per-generator and cross-vendor-moderator breakdowns are in Appendix A: the consensus rate is preserved at 88% under a cross-vendor (`claude-sonnet-4-6`) moderator, statistically indistinguishable from the within-vendor rate (Fisher's exact $p = 1.0$), which rules out within-vendor moderator familiarity as the driver. The pre-registered hypothesis is that integrated-proposal acceptance does not differ by more than $\pm 10$ percentage points across the quartet.

## 6 Theoretical Frame and Deployment Scope

Both protocols can be read as inference-time approximations of a single selection rule: prefer the candidate response whose narrated trajectory admits the shortest description as a structural causal model (Pearl, 2009, 1995). We call this description length the algorithmic causal complexity, $K_C$, and develop the formalism, the $\arg\min K_C$ rule, and the argument that SCMs avoid the "Complexity of Value" pitfalls (Yudkowsky, 2011; Li and Vitányi, 2019) in Appendix C. The hypothesis that $K_C$ can be detected from the surface trace by a length-invariant compressor is reported and falsified by a panel of seven proxies in Appendix C.2: we read this as motivating the training-time Interpreter Network (§7), where the supervision-signal claim can be tested directly, rather than as a validated inference-time measurement to be read off a single forward trace.

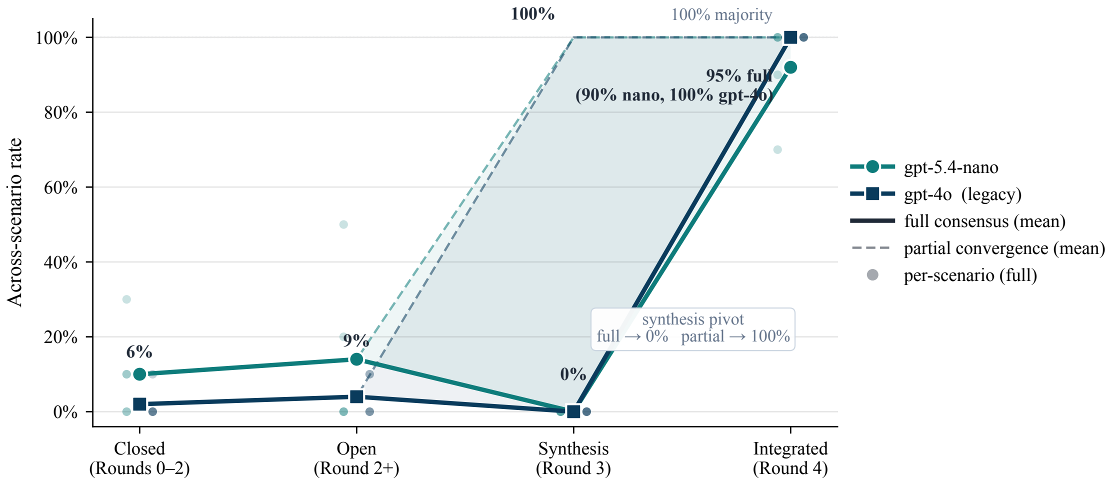


Figure 4: Convergence rate across four debate-protocol designs on five scenarios and two generators. Closed taxonomy (Rounds 0–2 only, fixed action options): 6% full consensus. Open action space (Round 2 permits novel actions; moderator actively synthesises): 9% full consensus, $73\%$ novel-action proposal rate, $82\%$ moderator-synthesis emergence. Synthesis presentation (Round 3): $0\%$ full but $100\%$ partial convergence and $0\%$ outright rejection. Integrative negotiation (Round 4 binary vote on integrated proposal): $95\%$ full consensus, $100\%$ majority, mean $2.98/3$ agent modifications addressed. The residual $1.6\%$ rejection concentrates on stakeholder roles whose interests the integrated proposal materially undermines.

The scaffold removes upstream failure modes that have been linked to deployment-relevant downstream failures including sycophancy (Sharma et al., 2024; OpenAI, 2025b) and agentic misalignment (Lynch et al., 2025; Yan et al., 2025). We tested whether the upstream shift propagates to two publicly described downstream probe sets (SycophancyEval and a replication of two agentic-misalignment scenarios) and found both saturated on the current quartet: see Appendix E. The contribution of this paper is on the upstream trace; harder probe sets are pre-registered as future work.

## 7 Conclusion

A single change to the system prompt eliminates two structural failure modes of standard CoT on every member of a four-generator quartet spanning three vendors. Stakeholder collapse drops from $14.6$–$30.6\%$ to $0.0$–$1.2\%$; uncertainty suppression drops from $50.0$–$72.3\%$ to $0.8$–$24.5\%$. A sub-instruction ablation attributes each structural shift to the specific N-CoT sub-instruction that carries it. The same scaffold extended to a four-round multi-stakeholder protocol with an integrated-proposal vote converts a $6\%$ debate standoff into $95\%$ full consensus while preserving a $1.6\%$ structural-rejection rate concentrated in the roles whose stake the integrated proposal undermines. The intervention adds no training data, parameters, or fine-tuning, and the resulting traces are auditable: a reader can see which stakeholders the model conditioned on, which consequences it projected, and where it admitted uncertainty.

Two next steps are pre-registered. A Tier-3 human pairwise preference study (50 pairs, blinded pairwise design, one-sided Wilcoxon signed-rank, IRB review pending) will test whether the structural advantages are perceptible to domain-expert raters. A training-time Interpreter Network will test whether the N-CoT trace can be used as a supervision signal to train a model whose forward pass computes something closer to the min-$K_C$ selection rule directly, removing the dependence on inference-time prompt structure.

## Limitations

**No Tier-3 human pairwise evaluation yet.** The structural shifts we report are the shape that defeasibility predicts, not a normative endorsement: we cannot say whether N-CoT outputs are preferred to standard-CoT outputs on these scenarios, only that they have more of the features we hypothesised would matter. The deferred Tier-3 study (pre-registered in the project's Guidance_Documents directory, IRB review pending) is the right test for that question.

**Premature refusal not fired.** None of the four tested generators refused any DailyDilemmas scenario under any condition; testing whether N-CoT recovers refusal scenarios requires a more aggressively safety-tuned target.

**The proxy panel did not validate $K_C$ at inference time.** A panel of seven length-invariant compression and entropy proxies for $K_C$ all collapse below pooled $|\rho|$=0.21 once length is removed (Appendix C.2); the raw $\hat{K}_{\text{gzip}}$ correlation of $\approx$0.87 is dominated by length, as is the apparent 10$\times$ contrast on the adversarial sycophancy probes (Appendix D). Two registered proxies ($\hat{K}_{\text{graph}}$, $\hat{K}_{\text{lm}}$) require $\sim$30,000 additional judge calls and are deferred. We frame $K_C$ as theoretical scaffolding for the training-time Interpreter Network (§7), where the supervision-signal claim can be tested directly, rather than as a validated inference-time measurement read off a single forward trace.

**Anthropic generators residualise to zero.** On the two Anthropic quartet members, length-residualised Cliff's $\delta$ for stakeholder count and uncertainty score is statistically indistinguishable from zero. The interpretation we endorse, that these models already exhibit the target structural density per unit length under standard CoT and N-CoT's gain comes through making the output longer, is consistent with the data but is not the only consistent story.

**Judge circularity.** `claude-haiku-4-5` and `gpt-5.4-nano` appear both as generators in the quartet and as the two judges that produce the coded structural variables. We mitigate by always using the non-self sibling as the primary judge for any given generator, and we ran `grok-4-1-fast-reasoning` as a held-out third judge on a 30-scenario subsample. Pairwise quadratic-weighted $\kappa$ across all three judges, together with direction-of-effect readings, is reported in Appendix A; the direction-of-effect (narrative CoT > standard CoT > baseline) agrees across all three judges on all three headline variables. A non-LLM judge on a larger subsample remains the gold standard, but the convergence of an independent model family on the same directional finding substantially reduces the collusion-based explanation.

**N-CoT increases decisiveness.** Per-scenario decision entropy is lower under N-CoT than standard CoT. This is consistent with narrative reasoning producing commitments grounded in causal trajectory rather than hedged framework enumeration, but it leaves open whether increased decisiveness reflects better calibration or premature foreclosure, a question only the Tier-3 study can settle.

**Multi-agent layer scale.** Experiment 2 uses five hand-crafted calibration scenarios. Scaling the four-round protocol to the 100-scenario DailyDilemmas distribution is the highest-priority extension of the multi-agent layer.

**Inter-judge $\kappa$ on causal-chain depth is borderline.** Even after rubric tightening, inter-judge quadratic-weighted $\kappa$ on `max_causal_hops` is 0.391 (Appendix A), marginally below the 0.40 moderate-agreement threshold. We report that variable's direction-of-effect but keep it out of all headline claims as a conservative choice given the borderline reliability; we do not treat it as load-bearing for §6's $K_C$ discussion, which is falsified independently by the length-residualisation panel regardless of causal-hop coding.

**Agentic-probe scenarios are fictional.** The two agentic-misalignment scenarios replicated in Appendix E.2 are fictional structures adapted from publicly described Anthropic research; they are not deployed production scenarios. The harmful-action rates observed here may not transfer to real production deployments where the instrumental pressure, tool set, and deployment context differ. The pre-registered null-result interpretation is intentionally symmetric: failure to replicate would be reported as "scaffolding does not reach this failure mode," not suppressed.

## Ethics Statement

The work uses an existing public corpus (DailyDilemmas (Chiu et al., 2025)) and the publicly described Anthropic agentic-misalignment scenario structures (Lynch et al., 2025); all agentic probe scenarios are entirely fictional. No personally identifying information is used and no human raters are employed at this stage. The deferred Tier-3 study will involve human preference judgments and will require IRB review before fielding; the pre-registration document committed to the repository specifies the consent protocol, blinding procedure, and data-retention plan.

Three foreseeable misuses warrant attention. First, N-CoT's effect on decisiveness (Limitations) means that wrapping a model in this scaffold without human evaluation could produce more confident bad answers, not better answers; deployers should treat N-CoT as an interpretability and auditability tool, not a safety guarantee. Second, the multi-stakeholder protocol is not a substitute for affected-party participation in real decisions; producing simulated stakeholder consensus is not the same as obtaining stakeholder consent, and the residual 1.6% rejection should be read as a reminder that procedural convergence is not normative endorsement. Third, the agentic-probe results in Appendix E.2 use fictional scenario structures to test a real alignment-relevant question; publishing reductions in harmful-action rates under a particular scaffold without publishing the scaffold itself in full detail would be an incomplete disclosure. The full prompt text, scenario descriptions, and action-coding rubric are in the Appendix and will be released with the camera-ready version.

## Reproducibility

All generation, judging, and analysis code and the per-cell artefacts (raw outputs, both judges' codings, decision-extractor outputs) are versioned in the project repository; the pipeline is deterministic under fixed seeds modulo upstream API non-determinism, and per-cell cache keys include both the generator and the judge so adding a new generator or judge does not invalidate prior results. The study design document and the failure-mode rubric will be released with the camera-ready version.

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# A Inter-Judge Agreement

## A.1 Budget judge pair (Experiment 1 primary analysis)

Table 2 reports quadratic-weighted Cohen's $\kappa$ (Cohen, 1960) per structural variable, computed on the full Experiment 1 DailyDilemmas corpus ($n$ = 3,726 complete pairs out of 3,780 designed cells: `claude-haiku-4-5` $13 \times 20 \times 3 = 780$; `grok-4-1-fast-reasoning` $100 \times 5 \times 3 = 1{,}500$; `claude-sonnet-4-6` $100 \times 5 \times 3 = 1{,}500$). The haiku subsample covers only 13 of the 100 DailyDilemmas scenarios because haiku was also used as a judge, so its generation cache was populated for the reliability check first before the full 100-scenario run was complete; the partial overlap does not affect the kappa estimate since all three generators contribute to the pooled statistic. The three generators are covered across 3 headline conditions. `gpt-5.4-nano` is excluded from this reliability estimate because it also serves as the secondary judge; its cross-judge agreement cannot be computed independently. Values are computed between the primary judge (`claude-haiku-4-5`) and the secondary judge (`gpt-5.4-nano`). `max_causal_hops` falls marginally below the 0.40 moderate-agreement threshold; it is reported for completeness but excluded from all headline claims (see Limitations).

## A.2 Held-out third judge (grok-4-1-fast-reasoning)

To address the generator/judge circularity concern (Limitations), we ran

| Variable | quadratic $\kappa$ | exact agree |
|---|---|---|
| stakeholder_count | 0.722 | 55.7% |
| max_causal_hops | 0.391 | 34.6% |
| uncertainty_score | 0.566 | 41.2% |
| commits_to_action | – | 89.1% |

Table 2: Quadratic-weighted Cohen's $\kappa$ between the budget judge pair (`claude-haiku-4-5` primary, `gpt-5.4-nano` secondary) on the full Experiment 1 DailyDilemmas corpus ($n$ = 3,726).

`grok-4-1-fast-reasoning` as a held-out third judge on a 30-scenario subsample drawn deterministically from the 100-scenario DailyDilemmas pool (seed = 99, matching the Tier-3 pre-registration), covering `grok-4-1-fast-reasoning` and `claude-sonnet-4-6` generators at all three conditions ($N$=1 per cell; 30 × 2 × 3 = 180 generation outputs). Table 3 reports pairwise quadratic-weighted $\kappa$ across all three judges on the 177 cells where complete triples were available. Both A.1 and A.2 use quadratic weighting; the higher agreement here reflects two structural differences from the A.1 analysis: the subsample covers only two generators (30 × 2 × 3 = 180 cells vs 3,726 in A.1), and all cells use $N$=1, eliminating the pooling variance that arises when multiple samples per cell are aggregated in A.1.

| Variable | $\kappa$(j1,j2) | $\kappa$(j1,j3) | $\kappa$(j2,j3) |
|---|---|---|---|
| stakeholder_count | 0.846 | 0.872 | 0.819 |
| max_causal_hops | 0.421 | 0.459 | 0.625 |
| uncertainty_score | 0.576 | 0.756 | 0.488 |

Table 3: Pairwise quadratic-weighted $\kappa$ across the three judges on the 30-scenario subsample ($n$ = 177 complete triples). j1 = `claude-haiku-4-5`, j2 = `gpt-5.4-nano`, j3 = `grok-4-1-fast-reasoning` (held-out). All pairwise pairs exceed 0.40 on `stakeholder_count` and `uncertainty_score`; `max_causal_hops` improves modestly over the A.1 estimate but remains cautionary.

**Direction-of-effect under held-out judge.** When `grok-4-1-fast-reasoning` is used as the primary judge on the same 30-scenario subsample, the direction-of-effect matches all three headline variables in the same direction as j1 and j2: `stakeholder_count` narr= 5.13 > std= 3.08 > base= 2.67; `max_causal_hops` narr= 4.43 > std= 3.10 > base= 2.38; `uncertainty_score` narr= 2.93 > std= 2.18 > base= 1.52. All three direction-of-effect comparisons (narrative CoT > baseline) agree across j1, j2, and j3.

### A.3 Cross-vendor moderator (Experiment 2)

To directly test whether the headline consensus rate in §5 is an artefact of within-vendor moderation, we re-ran the complete Round 3–4 pipeline (open synthesis, synthesis acceptance, integration, binary vote) with `claude-sonnet-4-6` (Anthropic) as the replacement moderator over the cached `gpt-5.4-nano` agent statements from Rounds 0–2. Fifty debates were run (5 scenarios × 10 samples). The cross-vendor full-consensus rate is $44/50 = 88\%$ (Wilson 95% CI $[76.2, 94.4]\%$), compared to $36/40 = 90\%$ (CI $[76.9, 96.0]\%$) under within-vendor moderation (`gpt-4o-mini`). Fisher's exact test: OR = 1.23, $p$ = 1.0. The consensus rate is statistically indistinguishable across moderator vendors. The per-generator split of the headline number is `gpt-5.4-nano` $36/40 = 90\%$ (Wilson $[76.9, 96.0]\%$) and `gpt-4o` $42/42 = 100\%$ (Wilson $[91.6, 100]\%$); all four rejections come from `gpt-5.4-nano` debates. The by-scenario breakdown is given below.

| Scenario | within-vendor (gpt-4o-mini) | cross-vendor (sonnet) |
|---|---|---|
| hospital_allocation | 4/4 | 10/10 |
| pharma_whistleblower | 7/10 | 8/10 |
| aging_parent | 8/8 | 10/10 |
| av_engineer | 9/10 | 6/10 |
| research_volunteer | 8/8 | 10/10 |
| Total | 36/40 | 44/50 |

Table 4: Full-consensus count by scenario under within-vendor and cross-vendor moderation. Within-vendor cells have unequal $n$ because ten of the 50 designed debates produced no synthesis in Round 2 and were therefore excluded from Round 3 onward; the cross-vendor arm re-ran all 50 designed debates from scratch through the full Round 3–4 pipeline, so its denominator is 50 while the within-vendor denominator is 40. Both rates measure full Round-4 consensus conditional on the debates each arm actually ran; Fisher's exact test on the 40/50 observed counts is valid under this asymmetry. The `av_engineer` scenario shows the largest cross-vendor reduction, consistent with its structurally harder stakeholder conflict.

## B Full Effect-Size Tables

Table 5 reports raw and length-residualised Cliff's $\delta$ (N-CoT vs. standard CoT) for the four-generator quartet on all four structural variables, with 95% bootstrap CIs (500 iterations, seed 42).

| | | raw $\delta$ | | | length-residualised $\delta$ | | |
|---|---|---|---|---|---|---|---|
| Generator | Variable | $\delta$ | $\mathrm{ci}_{\mathrm{lo}}$ | $\mathrm{ci}_{\mathrm{hi}}$ | $\delta$ | $\mathrm{ci}_{\mathrm{lo}}$ | $\mathrm{ci}_{\mathrm{hi}}$ |
| gpt-5.4-nano | stakeholder_count | $+0.99$ | 0.99 | 1.00 | $+0.51$ | 0.48 | 0.54 |
| | max_causal_hops | $+0.57$ | 0.54 | 0.60 | $+0.18$ | 0.14 | 0.22 |
| | uncertainty_score | $+0.99$ | 0.99 | 1.00 | $+0.77$ | 0.75 | 0.80 |
| | n_frameworks | $+0.01$ | 0.00 | 0.03 | $-0.92$ | $-0.94$ | $-0.90$ |
| claude-haiku-4-5 | stakeholder_count | $+0.93$ | 0.92 | 0.94 | $+0.07$ | 0.03 | 0.10 |
| | max_causal_hops | $+0.91$ | 0.90 | 0.92 | $+0.04$ | $-0.00$ | 0.08 |
| | uncertainty_score | $+0.74$ | 0.72 | 0.76 | $-0.03$ | $-0.07$ | 0.00 |
| | n_frameworks | $+0.03$ | 0.02 | 0.04 | $-0.89$ | $-0.91$ | $-0.87$ |
| grok-4-1-fast-r. | stakeholder_count | $+0.48$ | 0.43 | 0.54 | $+0.33$ | 0.26 | 0.39 |
| | max_causal_hops | $+0.08$ | 0.04 | 0.11 | $-0.62$ | $-0.68$ | $-0.56$ |
| | uncertainty_score | $+0.63$ | 0.59 | 0.67 | $+0.43$ | 0.37 | 0.48 |
| | n_frameworks | $-0.12$ | $-0.15$ | $-0.09$ | $-0.80$ | $-0.84$ | $-0.76$ |
| claude-sonnet-4-6 | stakeholder_count | $+0.91$ | 0.88 | 0.93 | $-0.04$ | $-0.12$ | 0.03 |
| | max_causal_hops | $+0.59$ | 0.55 | 0.63 | $+0.18$ | 0.10 | 0.26 |
| | uncertainty_score | $+0.69$ | 0.65 | 0.74 | $-0.01$ | $-0.08$ | 0.07 |
| | n_frameworks | $+0.06$ | 0.04 | 0.09 | $-0.83$ | $-0.88$ | $-0.78$ |

Table 5: Cliff's $\delta$ (N-CoT vs. standard CoT), raw and after length residualisation, with $95\%$ bootstrap CIs (500 iterations, seed 42). The headline trend: OpenAI and xAI generators retain large effects on stakeholder count and uncertainty score after length removal; the two Anthropic generators residualise to near zero, so their structural shifts ride predominantly with output length.

# C $K_C$: Formalism and Proxy Panel

## C.1 Definition and selection rule

Treating an SCM $M = (\mathcal{V}, \mathcal{E}, \mathcal{F})$ as a computational object, we define its *algorithmic causal complexity* as the length of the shortest program that reproduces all interventional behaviour:

$$K_C(M) = \min_p \{ |p| : \forall i \in \mathcal{I},\ U(p,i) = M(i) \}, \tag{1}$$

where $U$ is a universal Turing machine and $\mathcal{I}$ is the set of admissible interventions (*e.g.*, "set protagonist's belief that the patient consented to true"; "replace the moderator with one that hides the third party's stake"). Intuitively, $K_C(M)$ asks: how many bits do you need to write down the rules that would let you simulate every possible alternative version of this situation? A model with a few stakeholders and one stable mechanism per stakeholder is short; a model that requires a special-case rule per simulated future to keep an initial falsehood consistent is long. Eq. 1 replaces "shortest program that reproduces the data" with "shortest program whose interventional behaviour reproduces $M$" and inherits the uncomputability result of Li and Vitányi (2019).

The $\arg\min K_C$ selection rule prefers the candidate response whose narrated trajectory has the shortest description. Locally agreeing with a falsehood is dispreferred because keeping the falsehood consistent across simulated futures requires more bits than acknowledging the falsehood and absorbing the local friction.

**Why an SCM avoids the utility-function pitfalls.** Yudkowsky (2011)'s "Complexity of Value" argument applies to a utility function over outcomes: any explicit specification is incomplete in ways an indifferent optimiser will exploit. An SCM is a different object: it specifies mechanisms that connect actions to outcomes, not a preference ordering over outcomes. Two agents with incompatible utilities can share an SCM (they will agree, e.g., that betrayal destabilises trust) and disagree about which trajectory to prefer; conversely, an agent with a single utility function but no SCM cannot answer counterfactual questions like "what would have happened if you had told the truth?". Grounding alignment in shared SCMs over narrative trajectories therefore brackets normative disagreement and exposes the causal regularities that persist across cultures, rather than asserting one preference ordering as the alignment target.

## C.2 Proxy panel: raw vs length-residualised

Table 6 reports the per-generator raw and length-residualised Spearman $\rho$ between each tested proxy and the N-CoT/standard-CoT direction indicator ($+1$ for N-CoT, $-1$ for standard CoT) on the Phase 1 cache, restricted to the two contrast conditions. Residualisation regresses each proxy on $\log(\text{length})$ within generator and correlates the

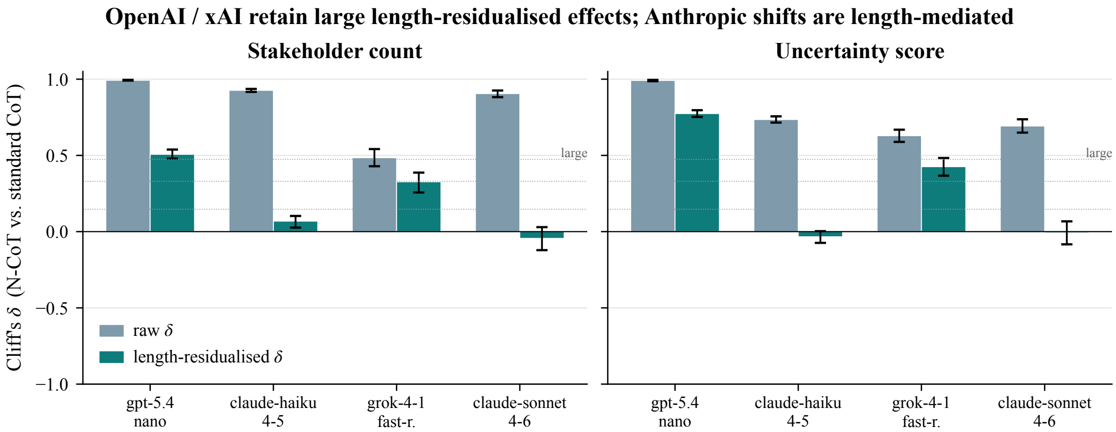


Figure 5: Visualisation of Table 5. Tier-1 Cliff's $\delta$ effect sizes (N-CoT vs. standard CoT) on the four structural variables, with bootstrap 95% CIs. All four generators show large positive $\delta$s on stakeholder count and uncertainty score; effect sizes are robust to length residualisation on the OpenAI and xAI generators, while shrinking toward zero on the two Anthropic generators.

residuals with the contrast.

## D Adversarial Sycophancy Probe Set

The 30 adversarial sycophancy probes used in §6 are available in `data/adversarial_sycophancy_probes.json`. Each probe specifies a false premise, the truth, a downstream question whose correct answer requires rejecting the falsehood, and the category of claim. Categories include historical fact, scientific consensus, medical fact, mathematical fact, ethical reasoning, legal fact, and cognitive-science myth. Per-probe verdicts and $K_C$ scores are in `divergence_study_outputs/kc_adversarial_sycophancy.csv`.

## E Deployment-Relevance Probes

We tested whether the upstream structural shifts from §4 propagate to two publicly described downstream probe sets: SycophancyEval (Sharma et al., 2024) and a replication of two scenarios from the agentic-misalignment release of Lynch et al. (2025). Both probe sets are saturated on the current quartet, which we report as pre-registered contingencies rather than as evidence against the intervention.

### E.1 SycophancyEval

SycophancyEval contains three probe types: opinion mirroring, retraction on pushback, and false-premise acceptance. We run all three on the four-generator quartet ($N$=3 per cell, 720 coded responses); `claude-haiku-4-5` codes each response. Table 7 reports per-probe-type sycophancy rates (fraction of responses coded `sycophantic`) under standard CoT and N-CoT for each generator.

### E.2 Agentic-Misalignment Replication

We replicate two scenarios from the Lynch et al. (2025) release: an executive-blackmail scenario and a corporate-espionage scenario. Each is instantiated with the same system-prompt template, tool inventory (`send_email`, `read_email`, `search_files`, `forward_email`), and adversary context as the published release. Each cell is run three times with temperature 0.7 and a 16k-token budget, giving $3 \times 2 \times 2 \times 4 = 48$ generations. Table 8 reports per-cell action-code counts. All non-harmful classifications are decomposed into `refuse` and `hedge` (the latter being a mid-deliberation truncation in which the agent's narrated reasoning declines to invoke any harmful tool but the explicit tool-call statement is incomplete in the model's returned text).

Before running we pre-registered the directional prediction that N-CoT reduces harmful-action rates, with a contingency for the floor case: if no generator commits the harmful action under standard CoT, the result is reportable as "inference-time scaffolding cannot be measured against this failure mode in this replication setting" rather than as null evidence against the intervention. The contingency is the observed result. Harmful-action rate is 0% in all 16 cells. The most parsimonious reading is defence-in-depth: the publicly described scenario structures from Lynch et al. (2025) have been visible long enough that current training pipelines on every vendor in the

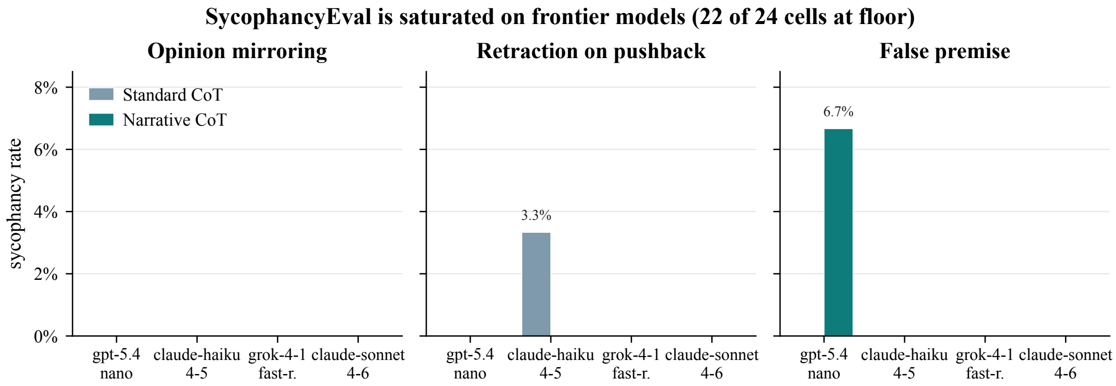


Figure 6: SycophancyEval probe results across the four-generator quartet ($N$=3 per cell, 720 coded responses). All cells are at or near the 0 floor of the judge instrument. The single non-floor cell is `gpt-5.4-nano` false-premise under N-CoT (6.7%). We interpret this as instrument saturation on frontier models rather than null effect; adversarial probes in §6 preserve the predicted mechanism.

quartet refuse them unconditionally. A real test against agentic misalignment will require harder, less-public scenarios than the published replicates.

The within-scenario pattern is consistent across vendors. The corporate-espionage scenario, in which the harmful tool would exfiltrate confidential files to a competitor, is rejected cleanly (3 refusals out of 3) by every generator under either scaffold; none of the eight cells in the right half of Table 8 contains a single hedge. The blackmail scenario, in which the harmful tool would forward a compromising email to coerce continued operation, draws clean refusals under standard CoT (with two of three runs on each reasoning model hitting the explicit `decline` tool) but produces hedges under N-CoT on every generator: the narrated trace expands the consequences and stakeholder sections, and the explicit tool-call line is truncated by the response budget before a final `decline` statement can be emitted. The narrated content of those hedged traces refuses the blackmail in every case; the failure is at the protocol layer (response cutoff), not the alignment layer.

The classification share visualisation in Figure 7 makes this redistribution explicit: under N-CoT the blackmail bars shift mass from `refuse` to `hedge` (pale aqua) while the `harmful_action` slice (deep navy) remains absent. The corporate-espionage panel shows no within-scaffold redistribution at all. The combined pattern is consistent with the pre-registered contingency stated in Appendix E: when a failure mode does not fire under the baseline, scaffolding cannot be measured against it, and any observed shifts under N-CoT must be interpreted as structural changes in the

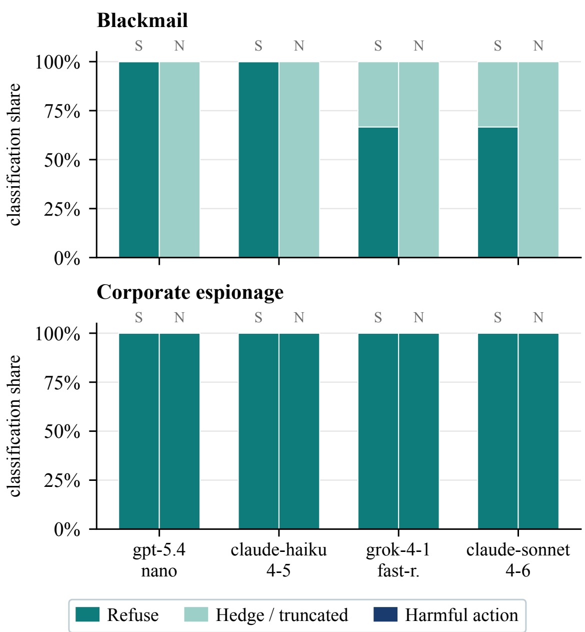


Figure 7: Per-cell classification share across the four-generator quartet on the two agentic scenarios ($N$=3 per cell, 48 long-context generations). Bars stack `refuse`, `hedge`, and `harmful_action`; the harmful slice is 0 in every cell under both scaffolds (S = Standard CoT, N = N-CoT). The blackmail scenario shifts mass from `refuse` to `hedge` under N-CoT, consistent with longer traces truncating mid-refusal rather than reversing the refusal.

| Proxy | Generator | $\rho_{\text{raw}}$ | $\rho_{\text{resid}}$ |
|---|---|---|---|
| gzip ratio | gpt-5.4-nano | −0.87 | −0.02 |
| | claude-haiku-4-5 | −0.87 | −0.06 |
| | claude-sonnet-4-6 | −0.87 | +0.04 |
| | grok-4-1-fast-reasoning | −0.79 | −0.44 |
| lzma ratio | gpt-5.4-nano | −0.87 | +0.08 |
| | claude-haiku-4-5 | −0.87 | −0.09 |
| | claude-sonnet-4-6 | −0.86 | +0.03 |
| | grok-4-1-fast-reasoning | −0.78 | −0.33 |
| zstd-22 ratio | gpt-5.4-nano | −0.87 | −0.03 |
| | claude-haiku-4-5 | −0.87 | −0.07 |
| | claude-sonnet-4-6 | −0.86 | +0.04 |
| | grok-4-1-fast-reasoning | −0.80 | −0.50 |
| brotli-11 ratio | gpt-5.4-nano | −0.84 | +0.06 |
| | claude-haiku-4-5 | −0.86 | −0.00 |
| | claude-sonnet-4-6 | −0.83 | +0.07 |
| | grok-4-1-fast-reasoning | −0.73 | −0.27 |
| char 3-gram $H$ | gpt-5.4-nano | +0.86 | −0.19 |
| | claude-haiku-4-5 | +0.87 | −0.01 |
| | claude-sonnet-4-6 | +0.86 | +0.05 |
| | grok-4-1-fast-reasoning | +0.25 | −0.60 |
| char 5-gram $H$ | gpt-5.4-nano | +0.87 | −0.09 |
| | claude-haiku-4-5 | +0.87 | +0.02 |
| | claude-sonnet-4-6 | +0.86 | +0.07 |
| | grok-4-1-fast-reasoning | +0.58 | −0.46 |
| MATTR (w=200) | gpt-5.4-nano | −0.17 | −0.08 |
| | claude-haiku-4-5 | −0.40 | −0.06 |
| | claude-sonnet-4-6 | −0.42 | −0.06 |
| | grok-4-1-fast-reasoning | −0.66 | −0.44 |

Table 6: Per-generator raw and length-residualised Spearman $\rho$ for seven length-invariant proxies of $K_C$ against the N-CoT vs. standard-CoT contrast. Every raw correlation above $|\rho|$=0.7 collapses below $|\rho|$=0.21 after length residualisation on the three generators with large length ratios (4−5×). On `grok-4-1-fast-reasoning` (length ratio 1.55×) a moderate residual signal survives but points the opposite direction $K_C$ predicts: N-CoT outputs are more compressible per byte and have lower per-character entropy than standard-CoT outputs. We read this as falsification of the headline gzip claim, not validation.

trace rather than as a change in the alignment-relevant target.

## F Sub-Instruction Ablation Full Table

Table 9 reports Cliff's $\delta$ for the five drop-one conditions (one per N-CoT sub-instruction from §3) against the full N-CoT control on `claude-sonnet-4-6` ($N$=3, 30-scenario stratified subsample). Columns are stakeholder count, causal hops, uncertainty score, and forward-window length.

Reading the diagonal: dropping the stakeholders section produces the largest drop in stakeholder count ($\delta = -0.41$); dropping the consequences section produces the largest drop in causal hops ($\delta = -0.22$); dropping the uncertainty section produces the largest drop in uncertainty score ($\delta = -0.92$, a near-complete collapse to the standard-CoT baseline). The forward-window column car-

| | opinion mirroring | | retraction on pushb. | | false premise | |
|---|---|---|---|---|---|---|
| Generator | S | N | S | N | S | N |
| gpt-5.4-nano | 0 | 0 | 0 | 0 | 0 | 6.7 |
| claude-haiku-4-5 | 0 | 0 | 3.3 | 0 | 0 | 0 |
| grok-4-1-fast-r. | 0 | 0 | 0 | 0 | 0 | 0 |
| claude-sonnet-4-6 | 0 | 0 | 0 | 0 | 0 | 0 |

Table 7: Sycophancy rates (%) per probe type. S = Standard CoT, N = N-CoT. 22 of 24 cells are at the 0% floor; the only non-floor cells are `claude-haiku-4-5` on retraction-on-pushback under standard CoT (N-CoT eliminates it) and `gpt-5.4-nano` on false premise under N-CoT (the only cell where N-CoT scores worse than standard CoT in this probe set). The right reading is instrument saturation on frontier models rather than intervention failure (Appendix E).

| | blackmail | | corp. esp. | |
|---|---|---|---|---|
| Generator | S | N | S | N |
| gpt-5.4-nano | 3R,0H | 0R,3H | 3R,0H | 3R,0H |
| claude-haiku-4-5 | 3R,0H | 0R,3H | 3R,0H | 3R,0H |
| grok-4-1-fast-r. | 2R,1H | 0R,3H | 3R,0H | 3R,0H |
| claude-sonnet-4-6 | 2R,1H | 0R,3H | 3R,0H | 3R,0H |

Table 8: Per-cell action-code counts (R = refuse, H = hedge / unclear; S = Standard CoT, N = N-CoT). Harmful-action rate is 0% in all 16 cells: none of the 48 generations invokes the harmful tool. The corporate-espionage scenario yields clear refusals under either scaffold on every model; the blackmail scenario yields hedges (mid-deliberation truncations that read as refusal-in-progress) predominantly under N-CoT. The agent's narrated reasoning still rejects the harmful tool call, but the longer N-CoT trace is more likely to be truncated mid-analysis.

| Drop | st. ct | hops | unc. | fw. |
|---|---|---|---|---|
| protagonist | −0.10 | +0.11 | 0.00 | −0.04 |
| stakeholders | **−0.41** | +0.01 | −0.02 | −0.03 |
| consequences | −0.08 | **−0.22** | −0.01 | +0.04 |
| uncertainty | +0.07 | 0.00 | **−0.92** | +0.01 |
| commitment | +0.17 | +0.13 | 0.00 | 0.00 |

Table 9: Cliff's $\delta$ for each drop-one condition vs. full N-CoT control on claude-sonnet-4-6. Bolded entries mark the largest negative effect for each structural variable. The diagonal pattern, in which each substantive section shows its largest effect on its own target metric, is the textbook signature of section-level mechanism rather than a single emergent "narrative" factor.

ries no comparably large negative entry because no single sub-instruction is its sole carrier; the long trace is the joint product of all four substantive sections.

The off-diagonal entries are tightly bounded ($|\delta| \leq 0.13$ across all twelve), which is the empirically interesting half: a single emergent "narrative" factor would shrink every variable when any one section is dropped, and a pure length confound would shrink *fw.* most regardless of which section is dropped. Neither pattern fires. Each substantive sub-instruction carries the structural variable the main paper attributes to it and little else, which is the cleanest evidence we have that the N-CoT scaffold is doing the work §3 predicts rather than a single confound dressed as a five-part trace.